A Multi-Component System for Multi-Label ICD-10 Classification of Greek Cardiology Discharge Summaries

BioASQ ElCardioCC at CLEF 2026

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Abstract

We present a multi-component system for the ElCardioCC 2026 shared task on automatic ICD-10 coding of Greek cardiology discharge summaries. Our approach builds and trains six complementary components from scratch: a fine-tuned Greek BERT model, two XLM-RoBERTa variants (base and large), a hybrid information retrieval module, a dictionary-based rule system, and a named entity recognition with entity linking pipeline. Individual models are trained with task-specific loss functions (Asymmetric Loss, ZLPR) and per-label threshold tuning on the validation set. Rather than selecting a single model, we fuse all six outputs through a declarative metaheuristic ensemble that searches over eleven base strategies and three composition operators via random restarts, hill-climbing, and Variable Neighbourhood Search (VNS). Our best submission achieves a micro-F1 of **0.8667** on the official test set, ranking first among all participating teams and improving by approximately 1.8 percentage points over the best standalone model.

Keywords

automatic clinical coding, multi-label classification, metaheuristic ensemble, Greek BERT, cardiology discharge summaries, low-resource NLP

1. Introduction

Clinical coding assigns standardised diagnostic codes to free-text medical records, a mandatory step in hospital administration, epidemiological surveillance, and insurance reimbursement. Manual coding is slow, costly, and prone to inter-annotator disagreement, motivating automatic approaches [5]. The ElCardioCC 2026 shared task (BioASQ/CLEF 2026) targets this problem for Greek cardiology discharge summaries, requiring systems to predict all applicable ICD-10 codes per document, a multi-label classification problem over 115 codes.

Three challenges define the setting. First, Greek is a morphologically rich, low-resource language for clinical NLP, with discharge summaries mixing Greek terminology and transliterated abbreviations (PCI, TAVI, CABG). Second, some summaries exceed the 512-token BERT limit. Third, the 115 ICD-10 codes span frequencies from a single instance to over 1,000.

No single architecture dominates across all labels: neural models excel on frequent labels, dictionary rules on procedural codes, retrieval on rare coverage, and NER+EL on mention-level disambiguation. We therefore build and train six complementary components, each targeting different failure modes, and fuse their outputs through a declarative metaheuristic ensemble that searches over 11 base strategies and 3 composition operators on the validation set.

CLEF 2026 Working Notes, 21–24 September 2026, Jena, Germany
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2. Background

Automatic ICD coding from clinical text has been addressed with CNN/RNN architectures [11], BERT-based models [4, 7], and hybrid retrieval-classification pipelines [5]. For Greek clinical NLP, Greek BERT [6] provides a domain-relevant starting point, while XLM-RoBERTa [1] handles Greek–Latin code-switching in clinical abbreviations. Label imbalance is the central challenge in multi-label coding: Asymmetric Loss (ASL) [10] addresses this by suppressing easy negatives more aggressively than positives. Ensemble methods [3] and VNS-based search [8] over non-differentiable objectives complement neural approaches for rare-label coverage.

3. Task and Dataset

3.1. Task Definition

ElCardioCC 2026 is a multi-label ICD-10 coding shared task for Greek cardiology discharge summaries (BioASQ/CLEF 2026), with 115 target codes.

3.1.1. Evaluation metric.

The primary metric is a group-level F1 score that reflects clinical equivalence among related codes. Each gold annotation is a *group*, a set of interchangeable ICD-10 codes. Scoring is defined as follows:

- TP: a gold group is matched if at least one of its member codes is predicted.
- FN: gold groups with no predicted member.
- FP: predicted codes that do not belong to any gold group.

Multiple predictions within the same group do not increase TP. Precision, recall, and F1 are then:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \quad R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \quad F1 = \frac{2PR}{P + R} \tag{1}$$

This differs from standard multi-label micro-F1: predicting any one equivalent code fully satisfies the corresponding group, whereas predicting codes outside any gold group is penalised as false positives.

Example. Given gold groups $\{\{\text{R55}\}, \{\text{I10, I11}\}, \{\text{I44}\}, \{\text{Z95}\}\}$:

- Predicting {R55, I10, I11, I44, Z95}: TP=4, FP=0, F1=1.0; both I10 and I11 satisfy the same group without penalty, since neither falls outside a gold group.
- Predicting {R55, I10, I44, Z95, I20}: TP=4, FP=1, F1=0.889; I20 does not belong to any gold group and is counted as a false positive.

3.2. Dataset

The dataset consists of 2,500 anonymised discharge summaries. We split these into training (2,000), validation (250), and test (250) sets; a separate blind test set is held by the organisers. Documents are moderately long: mean 280 tokens (≈2,000 characters), maximum 1,218 tokens. Only 2.4% of documents exceed the 512-token BERT limit.

The 115 codes are highly skewed. The five most frequent (Z95, I25, I21, I48, E11) account for ∼41% of all annotations; 30 codes appear fewer than 10 times and 4 codes appear only once (Table 1). This long-tailed distribution requires explicit strategies for rare-label coverage.

Table 1
Label frequency distribution across 115 ICD-10 codes.

Frequency band	# codes	Approx. % of instances
≥ 500	5	41%
100–499	22	36%
10–99	58	21%
< 10	30	<2%

3.3. Preprocessing and Data Augmentation

Text cleaning is shared across all splits: whitespace collapse and removal of non-essential special characters (preserving hyphens, commas, slashes, parentheses). Greek BERT additionally receives lowercasing and accent stripping; XLM-RoBERTa receives whitespace-only normalisation to preserve abbreviation signals ("PCI", "TAVI"). ICD-10 annotations are encoded as 115-dimensional binary vectors.

Stratified splitting uses a two-stage Multilabel Stratified Shuffle Split [12] to produce an 80/10/10 partition, ensuring every validation and test label also appears in training. Of the 2,500 samples, 1,453 include mention-level annotations enabling NER+EL training.

Synonym augmentation (XLM-RoBERTa base only) swaps dictionary terms with clinical synonyms at $p = 0.35$ using frequency-adaptive multipliers ($\times 4$ for 30–99 instances, $\times 3$ for 100–199, $\times 2$ for 200–299).

4. Methodology

The system pipeline is: (i) train six independent components; (ii) tune per-model decision thresholds on the validation set; (iii) search for the best ensemble fusion strategy. Figure 1 provides an overview.

Raw JSONL (train / validation / test)

↓

Preprocessing & Stratified Split

↓

Greek BERT	XLM-R Large	XLM-R Base	Hybrid IR	Dictionary Baseline	NER +EL

↓

Per-model threshold tuning (validation set)

↓

Metaheuristic Ensemble (11 strategies + 3 compositions)

↓

Final Predictions

Figure 1: High-level system pipeline.

4.1. Greek BERT Multi-Label Classifier

4.1.1. Architecture.

We fine-tune `nlpaueb/bert-base-greek-uncased-v1` (Greek BERT) [6] with a two-layer MLP head: mean+CLS concatenation pooling (1536-dim) $\rightarrow$ Dropout(0.2) $\rightarrow$ Linear(1536 $\rightarrow$ 768, GELU) $\rightarrow$ Dropout(0.2) $\rightarrow$ Linear(768$\rightarrow$115). Concatenating mean-pooled hidden states and the [CLS] vector outperformed plain CLS extraction by ∼0.8 percentage points in our ablation, as diagnostic evidence is distributed across the full document. Max sequence length: 256 tokens.

4.1.2. Training.

Table 2 lists the final hyperparameters. ASL [10] replaced binary cross-entropy (BCE) after Phase 2, raising validation micro-F1 from ≈0.74 to 0.788 by suppressing easy negatives. ASL applies asymmetric focusing to positive and negative terms:

$$\mathcal{L}_{\text{ASL}} = \sum_{y} \begin{cases} (1-p)^{\gamma^+} \log p & y = 1 \\ (p_m)^{\gamma^-} \log(1-p_m) & y = 0 \end{cases} \tag{2}$$

where $p_m = \max(p - m, 0)$ shifts and clips easy negatives, $\gamma^+ = 1$, $\gamma^- = 4$, $m = 0.05$. **LLRD** (factor 0.90) updates lower encoder layers at LR $\times\, 0.90^l$, preserving pre-trained Greek representations while adapting upper layers to the clinical domain. The model evolved across four phases: BCE baseline (0.74) → ASL (0.788) → LLRD + MLP (0.811 at t=0.5) → final P4 tuning (0.7984 at t=0.6, 0.8062 after per-label threshold tuning). The drop from 0.811 to 0.7984 reflects the stricter active threshold (t=0.6 vs t=0.5); per-label tuning then recovers and exceeds the prior peak.

Table 2
Greek BERT final hyperparameters.

Hyperparameter	Value
Max length / batch	256 tok / 32 (16 + grad. accum. 2)
Learning rate / LLRD	3×10^{-5} / 0.90 per layer
Warmup / weight decay	6% / 0.01
Loss	ASL ($\gamma^- = 4$, $\gamma^+ = 1$, clip $= 0.05$)
Epochs / patience	30 / 7 (BF16)
Pooling / head	mean+CLS concat / MLP 1536→768→115
Dropout	0.2
Active eval threshold	0.6

4.1.3. Threshold tuning.

A two-pass sweep (step 0.01) selects $t_l^* = \arg\max_{t\in[0.45,0.75]} F1_l(t; \mathcal{D}_{\text{val}})$ per label independently. Pass 2 accepts per-label values only for labels with ≥15 positive validation examples and ΔF1 > 0.0015. This adds ≈0.8 percentage points over the active threshold (t=0.6) and ≈3.7 percentage points over a fixed t=0.5 baseline, bringing the tuned validation micro-F1 to 0.8062.

4.2. XLM-RoBERTa Classifiers

XLM-RoBERTa's 250K multilingual vocabulary handles Greek–Latin code-switching better than Greek BERT's 35K vocabulary.

4.2.1. Long-document strategy.

A sliding window (stride 256) handles documents exceeding 512 tokens. At inference, chunk logits are fused as $\hat{y} = 0.5\,\bar{y}_{\text{mean}} + 0.5\,y_{\max}$, balancing average evidence with the most confident chunk signal.

4.2.2. Large variant.

ZLPR loss, effective batch 32 (batch 2, grad. accum. $\times 16$), LR 2×10^{-5}, weight decay 0.05, dropout 0.15, 40 epochs (patience 5). Validation F1 ceiling 0.7538; slower convergence and memory constraints (560M parameters) led us to focus development on Greek BERT.

4.2.3. Base variant.

Implements MedicalModelWithDescriptions: the `[CLS]` representation passes through 5 parallel dropout layers ($p = 0.1i$, $i \in \{1 \dots 5\}$) whose logits are averaged (multi-sample dropout). A semantic anchoring term adds cosine similarity between the document embedding and frozen ICD-10 description embeddings, scaled by a learnable scalar α:

$$\hat{y} = W\mathbf{h} + \alpha \cdot \tfrac{\mathbf{h}}{\|\mathbf{h}\|}\mathbf{D}^{\top} \tag{3}$$

where rows of $\mathbf{D}$ are unit-normalised (frozen MiniLM embeddings of Greek ICD-10 descriptions), so the term is a true cosine similarity. α converges from 0.099 to $\approx$0.12, confirming a consistent but small semantic correction. Training: AdamW LR 2×10^{-5}, early stopping (patience 4), per-class positive weights (ceiling 20). Post-processing: dynamic threshold tuning + ICD-10 hierarchy enforcement + co-occurrence rules. Validation micro-F1: **0.8101**. Despite using a smaller backbone, the Base model outperforms the Large variant (0.8101 vs. 0.7538). The primary factors are the semantic anchoring mechanism and task-specific post-processing absent in Large; GPU memory constraints (560M parameters) also forced micro-batch size 2 with gradient accumulation, limiting effective per-step gradient quality.

4.3. Information Retrieval Module

Three retrieval channels (BM25 [9], TF-IDF, dense MiniLM embeddings) are fused by Reciprocal Rank Fusion [2] ($\mathrm{RRF}(d) = \sum_i 1/(k + r_i(d))$, $k = 60$). Codes with RRF score $\geq 0.22 \times$ top score, capped at 12 per document, are predicted. The cap is critical: raising it to 25 collapses F1 from 0.662 to 0.140 (precision drops from 0.57 to 0.10 while recall rises modestly) because additional hits that pass the 0.22× score threshold are predominantly false positives.

4.4. Dictionary Baseline

A curated CSV (1,680 term-to-code entries) is matched by an Aho–Corasick automaton with a 35-term blacklist. Cardiology procedure rules fire on keywords such as PCI/PPCI/TAVI/CABG (adding Y84, Z95, and related codes); co-occurrence rules enforce clinical consistency (e.g. I22 $\Rightarrow$ I21); I25 $\Rightarrow$ Z95 is applied as a recall heuristic since most I25 patients in this corpus have vascular implants, though it introduces false positives corrected downstream by merge_and. Validation micro-F1 **0.7176** (precision 0.65, recall 0.80), covering 99.76% of documents.

4.5. Named Entity Recognition and Entity Linking

Greek BERT is fine-tuned as a BIO sequence tagger with a single entity type (`MED`) and a Partial CRF [13]. The coarse type maximises recall; the Partial CRF enforces label consistency via Viterbi decoding and enables learning from documents with only document-level annotations. Training: 6 epochs, batch 8, LR 2×10^{-5}, early stopping patience 2.

NER spans are augmented by Aho–Corasick dictionary matching; containment-based deduplication retains the longest span. Each mention m is linked to ICD-10 code c by:

$$\mathrm{score}(m, c) = 0.6 \cdot P_{\mathrm{prior}}(c \mid m) + 0.4 \cdot \cos(\mathbf{e}_{\mathrm{ctx}}, \mathbf{e}_c) \tag{4}$$

where P_{prior} is the (mention, code) co-occurrence frequency from training and $\mathbf{e}_{\mathrm{ctx}}$, $\mathbf{e}_c$ are MiniLM embeddings of a $\pm$200-char context window and the Greek ICD-10 description respectively. Full document-level dictionary boosting is applied after mention linking. Best validation micro-F1: **0.7223** at epoch 5.

4.6. Metaheuristic Ensemble

Rather than fixing a single fusion rule [3], we define a declarative strategy space in YAML and evaluate all strategies on the validation set, selecting the best-scoring configuration without code changes.

4.6.1. Base strategies.

Eleven base strategies cover the main fusion paradigms. The primary strategy is the weighted ensemble, a convex combination of all component probability vectors:

$$\hat{y} = \sum_{i=1}^{6} w_i\, \hat{y}^{(i)}, \quad w \in \mathcal{W} = \left\{ w \in \mathbb{R}^6 : \textstyle\sum_i w_i = 1,\ w_i \geq 0 \right\} \tag{5}$$

Additional strategies address complementary failure modes: majority vote pools predictions across all weight-search restarts (both classic and VNS); gated secondary falls back to IR/NER for documents where the weighted ensemble is uncertain; top-K pruning caps predictions per document; frequency-bucketed thresholding applies separate thresholds per label-frequency band; per-label champion routes each label to its highest-validation-F1 component; per-label champion + vote adds a label if the champion fires or a minimum number of other models agree; per-patient score routing selects the best-scoring model per document; and label correction mines co-occurrence rules ($a \to b$) from validation predictions, predicting b whenever a is predicted above a mined threshold.

4.6.2. Composition operators.

Three label-set algebra operators defined in `strategy_compositions.yaml` compose pairs of base strategy outputs $\{\hat{L}^{(s)}\}_{s\in\mathcal{S}}$, where $\hat{L}^{(s)} = \{l : \hat{y}_l^{(s)} \geq t_l\}$:

$$\hat{L}^{\text{or}} = \bigcup_{s\in\mathcal{S}} \hat{L}^{(s)} \qquad \text{(maximises recall)} \tag{6}$$

$$\hat{L}^{\text{and}} = \bigcap_{s\in\mathcal{S}} \hat{L}^{(s)} \qquad \text{(maximises precision)} \tag{7}$$

$$\hat{L}^{k\text{-of-}n} = \left\{ l : \sum_{s\in\mathcal{S}} \mathbb{1}[l \in \hat{L}^{(s)}] \geq k \right\} \qquad \text{(balances precision/recall)} \tag{8}$$

The best submission applies merge_and over the weighted ensemble and the label correction module, letting the correction step remove false positives from the ensemble output.

4.6.3. Weight optimisation.

Weights are optimised directly on the non-differentiable validation micro-F1 objective:

$$w^* = \underset{w\,\in\,\mathcal{W}}{\arg\max}\ \mathrm{F1}_\mu\left(\hat{y}(w),\ y^{(\text{val})}\right) \tag{9}$$

Two optimisers run independently, each for 2 restarts of 10,000 evaluations: (1) *classic*: random search over $\mathcal{W}$ followed by pairwise hill-climbing; (2) *VNS* [8]: shaking (radius-growing perturbation, $k_{\max}$=5) followed by a short hill-climb, with k reset on improvement and a random restart when all neighbourhoods are exhausted. The optimiser achieving higher validation micro-F1 provides the final weights; restart predictions from both are pooled for majority-vote strategies.

4.6.4. Submitted strategies.

The five submissions span the precision–recall trade-off: standalone Greek BERT (80/10 split and 100% data) establish the single-model ceiling; three ensemble submissions apply merge_and (precision-oriented), merge_k2 (balanced), and a safer-threshold merge_k2 variant. The merge_and composition achieves the best F1 by combining the weighted ensemble with the label correction module's precision boost.

5. Experimental Results

All micro-F1 values reported in this section use the group-level metric defined in Section 3: a gold group is satisfied by any one of its member codes, and over-predicting within a group counts as false positives.

5.1. Individual Component Performance

Figure 2 shows standalone group-level micro-F1 on our internal test split for each component. Greek BERT is the strongest single model (0.8196 on the internal test split), ahead of XLM-RoBERTa Base (0.8050). Dictionary and NER+EL perform comparably; the IR module has the lowest F1 but the highest recall (0.83), making it a key coverage contributor in the ensemble.

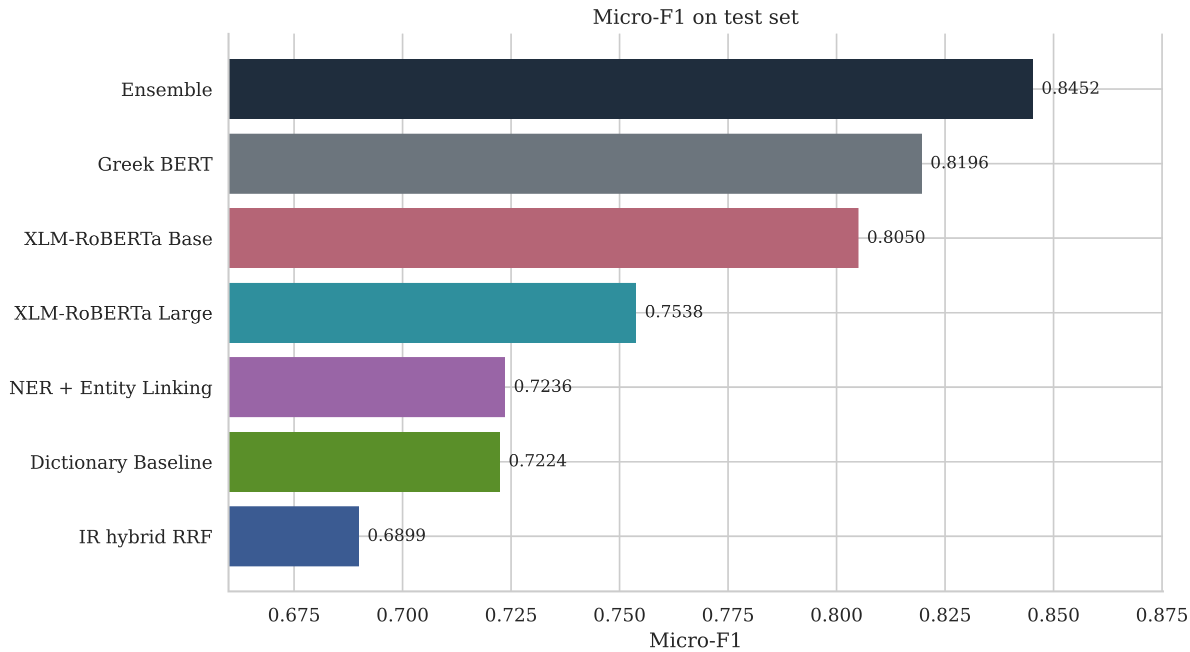

Figure 2: Standalone group-level micro-F1 per component on our internal 250-document test split. XLM-RoBERTa Large reports validation F1 (no separate test submission). Official test set scores appear in Table 5.

5.2. Ablation Studies

5.2.1. Threshold tuning.

Table 3 shows the effect of progressively finer thresholding. Per-label tuning adds ≈3.7 percentage points over a fixed t=0.5 baseline and ≈0.7 percentage points over the global optimum.

Table 3
Threshold tuning ablation for Greek BERT.

Strategy	Val. micro-F1
Fixed $t = 0.5$	0.7693
Fixed $t = 0.6$ (active)	0.7984
Global sweep $[0.45, 0.75]$	0.7989
Per-label sweep (2-pass)	0.8062

5.2.2. Impact of LLRD.

Layer-wise learning rate decay (factor 0.90) preserves pre-trained lower-layer representations while adapting upper layers to the clinical domain, most notably benefiting rare labels.

5.2.3. Loss function comparison.

ASL outperformed both binary cross-entropy and focal loss through better recall on rare labels. BCE gave good precision but lower recall; focal loss improved recall over BCE but less than ASL due to its symmetric treatment of positive and negative examples.

5.2.4. Ensemble composition.

Table 4 compares the composition operators. The intersection (merge_and) achieves the best F1 by combining the weighted ensemble with the label correction module's precision boost.

Table 4
Ensemble composition operator comparison (validation set).

Strategy	Recall	Precision	F1 trend
Union (merge_or)	highest	lowest	moderate
Weighted ensemble	mid	mid	strong
k-of-n (merge_k2)	mid	mid+	strong
Intersection (merge_and)	mid−	highest	best

5.3. Official Test Set Results

Table 5 reports all five submissions on the official test set. The best submission uses the merge_and strategy (weighted ensemble intersected with the label correction module).

Table 5
Official ElCardioCC 2026 test set results.

Submission	Recall	Precision	F1
ensemble (merge_and: weighted + correction)	0.8510	0.8830	**0.8667**
ensemble (merge_k2: weighted + per-label + corr.)	0.8687	0.8539	0.8613
ensemble (safer thr, merge_k2)	0.8767	0.8431	0.8596
mlc_greek_bert_100 (100% data)	0.8194	0.8811	0.8491
mlc_greek_bert (80/10/10 split)	0.8310	0.8675	0.8489

The ensemble outperforms the best standalone model by $+1.8$ percentage points (0.8667 vs. 0.8491), confirming the complementarity of the six components. The full-data Greek BERT achieves virtually the same F1 as the split version (0.8491 vs. 0.8489), suggesting the validation set contributes little additional signal at this scale.

The precision-oriented merge_and yields the highest overall F1 (0.8667), while merge_k2 variants achieve higher recall at the cost of precision, consistent with the 2-of-3 voting design. The safer-threshold merge_k2 reaches the highest recall (0.8767) but the lowest precision (0.8431).

5.4. Error Analysis

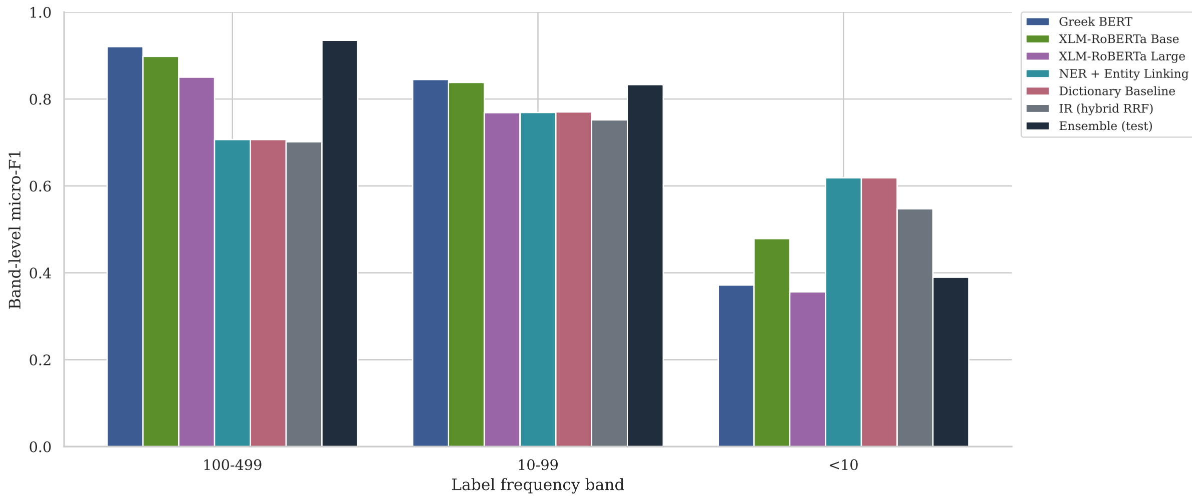

Figure 3: Macro-F1 per component across label frequency bands (three bands shown; no code reaches ≥500 instances in the 250-document validation set). All models degrade sharply on rare labels (<10 instances); the ensemble provides no recovery for this band.

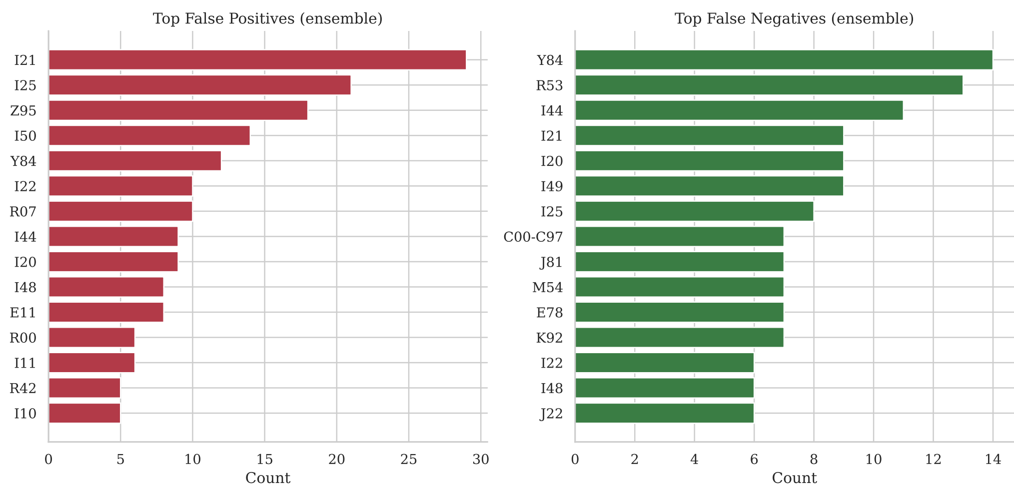

Figure 4: Top false-positive and false-negative codes for the best ensemble submission.

Rare labels. The 30 codes with <10 training instances are the primary error source. For 10 of these, per-label thresholding is not applicable. The dictionary recovers some (e.g. E85 via the amyloidosis rule), but codes with no lexical surface form (e.g. M35, R57) remain unrecovered.

Coverage gaps. 31 label types lie outside the dictionary's coverage, including metabolic (E00), haematological (D47), and musculoskeletal codes, which require inference from laboratory values or implicit multi-sentence evidence.

Label confusion. I21/I22/I25 (acute MI, subsequent MI, chronic ischaemic disease) frequently co-occur and require temporal reasoning to assign correctly, beyond current bag-of-token models.

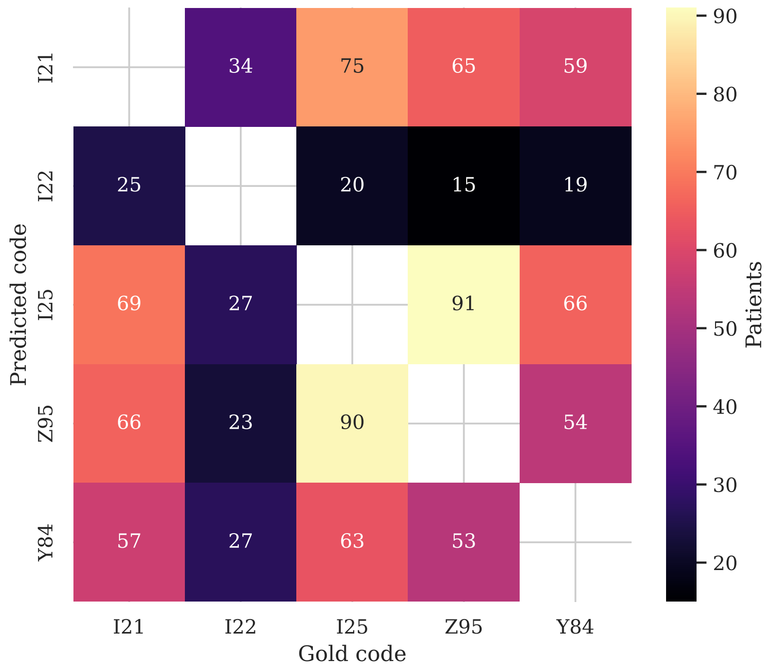

Figure 5: Co-prediction counts for the MI/procedure cluster (I21, I22, I25, Z95, Y84).

Z95 (vascular implants) is the main false-positive source due to ubiquitous stent and pacemaker mentions, motivating the merge_and correction step.

Calibration. Greek BERT under-predicts rare labels: probabilities for rare codes often fall below 0.5. Per-label thresholding helps but becomes unreliable for labels with fewer than 15 positive validation examples.

6. Conclusion

We presented a multi-component system for ElCardioCC 2026, achieving micro-F1 **0.8667** on the official test set and ranking first among all participating teams. The central finding is that paradigm diversity (neural classifiers, dictionary rules, retrieval, and NER+EL) matters more than tuning any single model, and that a metaheuristic search over fusion strategies can reliably exploit that diversity without manual engineering.

Several directions are worth exploring beyond the immediate fixes suggested by our error analysis. First, large language models (LLMs) have not yet been evaluated as ensemble components; few-shot prompting or retrieval-augmented generation could provide complementary coverage for rare and context-dependent codes. Second, the metaheuristic ensemble framework is task-agnostic: applying it to other clinical coding benchmarks (e.g. English, multilingual, or full ICD-10 hierarchies) would reveal whether the diversity benefit generalises or is specific to low-resource Greek. Third, the current pipeline treats each document independently; a session-level model that considers a patient's prior discharge summaries could resolve temporal ambiguities such as I21/I22/I25 that are intractable with bag-of-token representations. Finally, deploying such a system in a real hospital workflow raises open questions around active learning, human-in-the-loop correction, and concept drift as coding guidelines evolve over time.

Acknowledgments

This work was developed as part of the MSc in Artificial Intelligence and Data Analytics at the University of Macedonia.

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